LessonPlan Career Roadmap

Science - PCB (Physics, Chemistry, Biology): Class 6 to Career

LESSONPLAN CAREER ROADMAP
Compiled from publicly available sources. Items marked † should be confirmed on the official notification of the conducting body. Dates and rules change every year.

1. Pathway overview

Stage	When	Key decision	What it unlocks
Foundation	Class 6-8	Start Olympiads; build Science and Maths basics	Olympiad ladder, NMMS scholarship (Class 8)
Pre-board	Class 9-10	Maths level (CBSE Basic vs Standard); Junior Science Olympiad	Keeps PCMB open if Standard Maths is taken
Stream choice	After Class 10	Choose PCB (+ optional Maths / Computer Science / Psychology) in Class 11-12 (PUC in Karnataka)	Eligibility for NEET, CUET, IISER, life-science degrees
Preparation	Class 11-12	Boards + entrance exam focus (NEET / KCET / CUET / IAT)	Merit seats, scholarships (INSPIRE-SHE, CSSS)
Degree	After Class 12	Medical, allied health, pharmacy, life sciences, agriculture, research	Professional licence or research career
Postgraduate	3-6 years later	Specialise (MD/MS, MSc, M.Pharm, PhD)	Specialist and research roles
Career	-	Clinical, research, industry, public health, teaching	See section 7

2. Stage-by-stage nodes

Class 6-8: Foundation

- Decision: join Science, Maths and English Olympiads for exposure. Not mandatory, but it builds the profile used later.
- Unlocks: NMMS scholarship exam in Class 8†; INSPIRE-MANAK idea awards (Class 6-10).

Class 9-10: Pre-board

- Decision: in CBSE, choose Standard or Basic Maths for Class 10. PCB itself does not need Standard Maths, but Basic Maths closes the PCMB/Engineering option later. Take Standard if undecided. CBSE relaxed the rule for 2023-24 and 2025-26 (Basic Maths students may take Maths in Class 11 if the school agrees) but it may end with the new scheme.†
- Junior Science Olympiad (NSEJS) is for Class 10 and below.
- Prerequisite for PCB: Science in Class 10 with good marks in Biology and Chemistry concepts.

After Class 10: Stream choice

- Choose PCB. Add one optional subject: Maths (PCMB, keeps engineering and biotechnology open), Computer Science (bioinformatics), Psychology, or Physical Education.
- Karnataka note: Class 11-12 is the 2-year Pre-University Course (PUC) after SSLC; CBSE/ICSE schools run Class 11-12 directly.
- NEET requires Physics, Chemistry, Biology (or Biotechnology) and English in Class 12.

Class 11-12: Preparation

- Board marks matter: minimum aggregate in PCB is part of NEET eligibility, and KCET ranking uses PUC marks.
- Apply in parallel: NEET-UG, KCET (Karnataka), CUET-UG (central universities), IISER Aptitude Test, NEST.

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After Class 12: Degree

- Medical and related: MBBS, BDS, BAMS, BHMS, BUMS, BVSc & AH (all through NEET).
- Allied: Nursing, Pharmacy, Physiotherapy, Medical Lab Technology, Optometry, Radiology.
- Science and research: BSc Life Sciences, Biotechnology, Microbiology, Genetics, Biochemistry, IISER BS-MS.
- Agriculture and environment: BSc Agriculture, Horticulture, Forestry, Fisheries.

3. Olympiads and early exams (Class 6-10)

Exam	Conducted by	Classes	Levels	Tentative timing
National Science Olympiad (NSO)	Science Olympiad Foundation (SOF)	1-12	Level 1 (school) then Level 2 (top performers)†	Level 1 Oct-Dec, Level 2 Jan-Mar†
International Mathematics Olympiad (IMO)	SOF	1-12	Level 1, Level 2	Similar to NSO†
NSEJS then INJSO (Junior Science)	IAPT then HBCSE	10 and below (age rule applies)	Stage 1 NSEJS, Stage 2 INJSO, then selection camp for IJSO†	Stage 1 Nov, Stage 2 Jan-Feb†
NSEB then INBO (Biology)	IAPT then HBCSE	11-12	Stage 1 NSEB, Stage 2 INBO, then camp for IBO†	Stage 1 Nov, Stage 2 Jan-Feb†
INSPIRE-MANAK	Dept. of Science and Technology	6-12 (Class 11-12 added; Science stream only for those)	School nominates up to 5 ideas online; district then state then national	Last cycle nominations closed 15 Sep 2025; check inspireawards-dst.gov.in for next cycle
NMMS (scholarship test)	Karnataka: DSERT/KSEAB under Govt. of India scheme	8 (govt, aided and local body schools)	Single written test	Applications Sep-Oct; exam Jan-Feb (last held 1 Feb 2026); 2026-27 date not announced yet
NTSE	NCERT; Karnataka Stage 1 via DSERT	10	Stage 1 state, Stage 2 national	No sign it was discontinued, but sources are thin; 2026-27 notification expected Aug-Oct. Confirm at dsert.kar.nic.in and NCERT†

Tip for parents: Olympiads do not decide admission, but they sharpen concepts and give a ladder (school, state, national, international) that fits a Class 6 start.

4. Entrance exams after Class 10 / 12

NEET-UG (National Eligibility cum Entrance Test)	
Conducting body	National Testing Agency (NTA); official site neet.nta.nic.in
Used for	MBBS, BDS, BAMS, BHMS, BUMS, BVSc & AH, and several nursing and allied courses; AIIMS and JIPMER seats also use NEET
Minimum qualification	Class 12 with Physics, Chemistry, Biology/Biotechnology and English. Minimum aggregate in PCB: 50% General, 40% SC/ST/OBC, 45% PwBD (English not counted). The 2026 bulletin wording on this is reported unclear, and state AYUSH rules can differ.†
Age	Minimum 17 years on 31 Dec of the admission year (31 Dec 2026 for the 2026 cycle). No upper age limit is mentioned in 2026 coverage.
Frequency	Once a year
Tentative month	Registration Feb-Mar; exam normally first Sunday of May. 2026: held 3 May, cancelled 12 May after a paper leak, re-exam scheduled 21 June 2026. Dates can shift.†

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Pattern	180 compulsory MCQs: Physics 45, Chemistry 45, Botany 45, Zoology 45. Total 720 marks. Pen and paper (OMR). Duration reported as 180, 195 or 200 minutes by different sources.†
Marking	+4 for correct, -1 for wrong, 0 for unattempted
Languages	13 languages including English, Hindi and Kannada; language chosen at application is final

KCET (Karnataka Common Entrance Test)	
Conducting body	Karnataka Examinations Authority (KEA); portals kea.kar.nic.in and cetonline.karnataka.gov.in
Used for	Karnataka seats in B.Pharm, BSc Agriculture, Veterinary-allied, Farm Science, Engineering, Architecture and others. Medical, dental and AYUSH seats in Karnataka use the NEET score with KEA counselling.
Minimum qualification	Class 12/2nd PUC with Physics, Chemistry and Maths/Biology; percentage criteria (commonly 45% General, 40% reserved) and domicile/Kannada rules are in the KEA brochure†
Frequency	Once a year
Tentative month	April. 2026: 23-24 April, with a Kannada language test on 22 April for eligible (Horanadu/Gadinadu Kannadiga) candidates
Pattern	Each candidate takes three papers: Physics, Chemistry and either Biology or Maths. 60 MCQs per paper (180 in total), 1 mark each, 80 minutes per paper. Offline OMR, English and Kannada.
Marking	No negative marking
Rank basis	Combination of KCET score and 2nd PUC marks (commonly 50:50)†

CUET-UG (Common University Entrance Test)	
Conducting body	NTA, cuet.nta.nic.in†
Used for	BSc and other UG programmes at Central Universities and many state/private universities; ICAR agriculture UG admissions also use CUET†
Minimum qualification	Class 12 pass; each university sets its own subject requirement
Frequency / month	Once a year; May-June†
Pattern	Computer-based. Domain papers include Biology, Chemistry, Physics, Psychology: 50 MCQs per subject, all to be attempted, 60 minutes each. Up to 5 subjects in total including languages and General Test; 250 marks per subject (50 questions at 5 marks).†
Marking	+5 correct, -1 wrong, 0 unattempted

IISER Aptitude Test (IAT)	
Conducting body	IISER consortium, iiseradmission.in†
Used for	5-year BS-MS at all seven IISERs; 4-year BS at IISER Bhopal and Tirupati; 4-year BTech at IISER Bhopal
Minimum qualification	Class 12 Science with at least three of Physics, Chemistry, Biology, Maths (Biology not compulsory); minimum 60% aggregate, 55% for SC/ST/PwD
Frequency / month	Once a year; 2026 exam was 7 June (registration closed 13 April)
Pattern	Computer-based, 3 hours, 60 MCQs (15 each: Biology, Chemistry, Maths, Physics), 240 marks
Marking	+4 correct, -1 wrong, 0 unattempted

NEST (National Entrance Screening Test)

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Conducting body	NISER and UM-DAE CEBS
Used for	5-year integrated MSc in Biology and other sciences
Minimum qualification	Class 12 Science with minimum marks (about 60% General)†
Frequency / month	Once a year; typically June†
Pattern	Computer-based; sections on general, biology, chemistry, physics, maths†

Other routes (short list)

Note: NEET-UG 2026 was disrupted by a paper-leak cancellation, so applicants should follow the NTA and counselling announcements closely each year.

Exam / route	For	Note
AIIMS BSc Nursing entrance	Nursing at AIIMS institutes	Check whether replaced by NEET/CUET in current year†
BITSAT	B.Pharm at BITS Pilani	PCB route allowed for B.Pharm†
KCET / RGUHS counselling	BSc Nursing and paramedical in Karnataka	KCET (KEA) per an April 2023 government order; NEET reported as not required in Karnataka; 45% PCB General / 40% SC/ST, age 17 by 31 Dec; deemed universities (Manipal, Yenepoya) run their own tests. See Health and Allied document.†
Armed Forces Medical (AFMC)	MBBS via armed forces	Uses NEET, then AFMC selection

Postgraduate and beyond (preview)

Exam	Body	For	Tentative timing
NEET-PG	NBEMS	MD/MS after MBBS	Once a year; usually mid-year†
NEET-MDS	NBEMS	MDS after BDS	Once a year†
GATE (Biotechnology / Life Sciences)	IIT-organised	MTech, PhD, PSU jobs	Feb†
IIT JAM (Biological Sciences / Biotechnology)	IIT-organised	MSc at IITs/IISc	Feb†
CSIR-UGC NET (Life Sciences)	NTA	JRF, lectureship	Twice a year†
GPAT	NTA	M.Pharm	Once a year†
CUET-PG	NTA	MSc at central universities	Once a year†

5. Scholarships

Scholarship	Provider	Who can apply	Amount	Window
NMMS	Govt. of India via Karnataka DSERT	Class 8 in govt/aided/local body schools; family income up to 3.5 lakh; at least 55% in Class 7 (50% SC/ST)†	12,000 per year (1,000 per month), Class 9-12	Apply Sep-Oct; exam Jan-Feb
INSPIRE-MANAK	DST with NIF	Class 6-12, idea nominated by school (max 5 per school)	10,000 one-time	Nominations around Jul-Sep; portal inspireawards-dst.gov.in

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Scholarship	Provider	Who can apply	Amount	Window
INSPIRE-SHE	DST	Top 1% in Class 12 boards, or top 10,000 in JEE/NEET, or KVPY/NTSE/JBNSTS/Olympiad scholars, studying BSc / BS / Int. MSc in natural and basic sciences. No income limit.	80,000 per year (60,000 scholarship + 20,000 summer-project mentorship), renewable	Expected deadline 1 Dec 2026, not officially notified yet; portal online-inspire.gov.in†
CSSS (Central Sector Scheme)	Ministry of Education via National Scholarship Portal	Top 20% (above 80th percentile) of successful candidates in own board and stream; family income up to 4.5 lakh	12,000 per year for UG years 1-3; 20,000 per year for years 4-5 / PG	2026-27 portal opened 1 Jun 2026; one listing shows 31 Oct 2026 deadline†
Post-matric scholarships	National Scholarship Portal and Karnataka SSP	SC/ST/OBC/minority categories, income limits	Varies	Typically Jul-Nov†
Corporate / foundation scholarships	Private foundations	Merit and need based	Varies	Varies by foundation†

Note: KVPY was discontinued and its role absorbed by INSPIRE-SHE and NEET/JEE based selection. Confirm before listing it anywhere.†

6. Courses and colleges

Course	Duration	Entry	Note
MBBS	4.5 yr + 1 yr internship	NEET-UG	Gateway to doctor career and MD/MS
BDS (Dental)	4 yr + 1 yr internship	NEET-UG	Dentist; MDS later
BAMS / BHMS / BUMS	4.5 yr + 1 yr internship	NEET-UG; AACCC (15%) + state counselling	AYUSH systems; see Health and Allied document
BVSc & AH	About 5 yr	NEET-UG merit (VCI)	Veterinary; 50% PCB (47.5% reserved); see Agriculture document
BSc Nursing	4 yr	KCET + KEA counselling in Karnataka†	Hospital and community nursing; see Health and Allied document
B.Pharm / Pharm.D	4 yr / 6 yr	KCET, BITSAT, state CETs	Pharma industry, hospital pharmacy
BPT (Physiotherapy)	4.5 yr	State or NEET†	Rehabilitation, sports
BSc Allied Health (MLT, Radiology, OT, Optometry)	3-4 yr	NEET reported mandatory from 2026-27 (NCAHP)†	Diagnostics and technician roles
BSc Life Sciences / Biotechnology / Microbiology / Genetics	3-4 yr	CUET-UG or university	Research and industry route
BS-MS (IISER / NISER)	5 yr	IAT, NEST	Research focused
BSc Agriculture / Horticulture / Forestry	4 yr	CUET-UG (ICAR), KCET	Agri-business, research
BTech Biotechnology / Biomedical Engineering	4 yr	JEE / KCET	Needs Maths in Class 12 (PCMB only)

Examples of institutions (illustrative, not a ranking): AIIMS and JIPMER, Bangalore Medical College, St John's Medical College, Manipal and Kasturba Medical College, IISc, IISERs, NISER, Christ University, Mount Carmel College, RGUHS-affiliated colleges.†

7. Career outcomes

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Path	Job roles	Higher study
Medicine	Doctor, surgeon, specialist, public-health officer, medical officer in defence	MD/MS, DM/MCh, fellowships
Dental / AYUSH / Veterinary	Dentist, ayurvedic or homeopathic physician, veterinarian	MDS, MD (AYUSH), MVSc
Nursing / Allied health	Nurse, physiotherapist, lab technologist, radiographer, optometrist	MSc, MPT, hospital management
Pharmacy	Pharmacist, drug regulatory, clinical research, pharma industry	M.Pharm, MBA, PhD
Life sciences / Biotech	Researcher, lab scientist, bioinformatician, quality analyst	MSc, PhD, CSIR/ICMR fellowships
Agriculture / Environment	Agri-officer, plant breeder, forest service, food technologist	MSc Agri, PhD, UPSC/State services

8. Related umbrellas this stream connects to

- Design: biomedical and product design (NID DAT and UCEED accept any stream; confirm current eligibility).†
- Law: CLAT is open to Science students; medical law and forensic careers combine both.
- Defence: Armed Forces Medical Services and Military Nursing Service through NEET-linked selection; NDA Army wing allows any stream but Air Force/Navy need Maths and Physics.†
- Psychology and counselling: CUET-UG with Psychology; PCB students are eligible.
- Public policy and civil services: any graduate can attempt UPSC; Science graduates often choose Medical Science or Zoology as optional subject.†

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ONE-PAGE GIST: Science - PCB

Items marked † are subject to confirmation against official notifications.

Stage	What to do
Class 6-8	Try Olympiads (NSO/IMO); Class 8: NMMS scholarship test
Class 9-10	Take Standard Maths if unsure; Junior Science Olympiad (NSEJS)
After Class 10	Choose PCB (+ Maths or Computer Science optional); PUC in Karnataka
Class 11-12	Boards + NEET-UG; also KCET, CUET-UG, IAT, NEST
After Class 12	MBBS/BDS/AYUSH/Vet via NEET; Nursing, Pharmacy, Allied, Life Sciences, Agriculture, IISER
Postgraduate	NEET-PG, NEET-MDS, GATE, JAM, CSIR-NET, GPAT

Exam	When	Pattern	Marking
NEET-UG	Once a year, May (2026 re-exam: 21 Jun)	180 Q (Phy 45, Chem 45, Bot 45, Zoo 45); 720 marks; about 3 hr 20 min	+4 / -1
KCET	Once a year, April	Phy, Chem + Maths or Bio: 60 Q each, 80 min each	+1 / 0
CUET-UG	Once a year, May-June†	Up to 5 subjects, 50 MCQs each, 60 min	+5 / -1
IAT (IISER)	Once a year, June	CBT, 3 hr, 60 Q (15 each: Bio, Chem, Maths, Phy), 240 marks	+4 / -1
NEST	Once a year, June	CBT; multi-section science test	Check notification

Eligibility for NEET: Class 12 PCB + English, minimum 50% aggregate General (40% reserved), age 17+ by 31 Dec.†

Scholarships: NMMS (Class 8, 12,000/yr), INSPIRE-MANAK (Class 6-12, 10,000), INSPIRE-SHE (80,000/yr) and CSSS (12,000/yr) after Class 12, post-matric schemes via NSP/SSP.†

Careers: doctor, dentist, AYUSH physician, vet, nurse, physiotherapist, pharmacist, lab scientist, biotech researcher, agri-officer, public health.

Want a personalised roadmap or counselling? Contact LessonPlan, Yelahanka, Bangalore.

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